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A smart contract system for security of payment of construction contracts

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ABSTRACT

Timely payment of project participants is crucial for successful completion of construction projects, however the construction industry worldwide suffers from poor payment practices. The existing research studied the reasons and the consequences of the payment problem, but very few studies proposed methods for eliminating the problem. In this paper a novel smart contract payment security system named SMTSEC is presented for eliminating or reducing payment issues in the construction sector. The SMTSEC ensures security of payment of construction contracts through an automated computerized protocol that runs on a decentralized blockchain. The potential contributions and limitations of the proposed SMTSEC are studied through a real construction project. The main contribution of the SMTSEC is that it provides a new method for timely and transparent payment of construction projects by guaranteeing security of payment for works under construction without requiring administrative costs and burdens of trusted intermediaries such as lawyers or banks.

1. Introduction

In construction projects, a progress payment is a partial payment that covers the amount of work that has been completed during a period. The progress payments are in general made through a cascade system; the employer (as defined in [1]) to main contractor, main contractor to the subcontractors, and so on down the chain [2]. Due to its cascade nature, the traditional progress payment system may lead to major problems if payments are not made or made late.

Despite the fact that payments are the lifeblood of construction projects in practice steady fund flows are rare [3]. Payment problems lead to undesirable consequences including delays, increased costs, reduced performance, disputes, and bankruptcies [4,5] which could threaten the success of the projects. Payment problems are identified as one of the top dispute causes for construction projects [3,6,7].

Late payments by the employer impact contractor's cash flow negatively, which would also often lead to the payments of subcontractors and suppliers to be also deferred. Even if the contractor is paid on time, the prompt flow of payment down to the subcontractors is not always guaranteed because the contractor may use the payment to finance other projects, or may attempt to have additional profit by earning interest. Poor payment practices not only impact the majority of project participants' financial conditions destructively, but the trust between parties can also be severely damaged under certain circumstances [8]. Hence, the construction industry suffers from various payment issues. Therefore, there is an increasing need for a reliable and efficient progress payment system for resolving the payment issues and achieving successful projects.

Project Bank Accounts (PBAs) were proposed in recent years, to improve the traditional progress payment system by providing security of payment in the supply chain, and by decreasing the length of progress payment cycle. A PBA is a ring-fenced bank account from which payments are made directly and simultaneously to the main contractor and members of the supply chain [9]. Under PBA, the employer may deposit the entire project lump sum amount into the PBA or may pay amounts due once the progress payment reports are approved [10]. Upon issue of an authorization by the employer, due amounts are released from the PBA to the main contractor and subcontractors according to the breakdown included in the progress payment reports. The advantages of PBAs include accelerated payments, savings as a result of reduced financing and debt chasing expenses, and protection in the event of contractor insolvency [11]. Despite its advantages, PBAs

Abbreviations: APP, Actual Progress Payment Amount; Bloc, The Amount to be Blocked; Cong, Contingency Amount for the Blocked Funds; CryT, Cryptocurrency Type; Cur, Fiat Currency Type; DApp, Decentralized Application; EM, Employer; ETH, Ethereum; MC, Main Contractor; PayMC, Progress Payment Amount of the Main Contractor; Pay_i, Progress Payment Amounts to the Subcontractors and Suppliers; PBA, Project Bank Account; PeriBloc, The Period that the Employer's Funds Will Be Blocked; PeriPay, The Period that the Employer's Blocked Funds Can Only Be Used for Progress Payments; PoW, Proof-of-Work; PoS, Proof-of-Stake; PPP, Projected Progress Payment Amount; $SubPer_i$, Percentage Amounts of Payments that Will Be Paid Directly by the Employer to the Subcontractors and Suppliers; WAdEM, Employer's Wallet Address; WadMC, Main Contractor's Wallet Address; $WadSC_i$, Subcontractors' and Suppliers' Wallet Addresses

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have been only used in public projects mainly in the United Kingdom and Australia by government enforcement. Set-up and administration burdens and costs are often cited as the key barrier [10,12,13] for adaptation of the PBAs.

The recent developments in blockchain technology have led to the discovery of smart contracts as a creative method for automated execution of contract conditions. A blockchain is a method of decentralized consensus that was proposed to record transactions across a peer-to-peer network of computers. Blockchain was originally introduced for Bitcoin (a decentralized digital currency, or also referred as cryptocurrency), but then has evolved to other decentralized applications; such as smart contracts [14]. A smart contract is a computerized protocol whose implementation is automated through an executable code that runs on the blockchain. Smart contracts have control over the physical or digital objects according to the agreed terms [15]. The potential of smart contracts for contract management including their capability for the security of payments was mentioned in recent research [7,16–19]. However, very limited research efforts have been devoted for development of smart contract systems for security of payments, which is the main focus of this paper.

The main objective of this paper is to present a smart contract system for improving the progress payment process in the construction sector by providing an effective method for the secure, timely, and transparent payment of construction projects. The new smart contract payment method aims to enable a cost-effective and efficient alternative for security of payment of construction contracts. The rest of the paper is organized as follows: In section two, a review of research focusing on payment issues in the construction industry is provided along with the review of blockchain and smart contract studies in construction management. A brief summary of blockchain and smart contract technologies are included in section three, the research method is identified in section four, and the proposed smart contract-based security of payment system is presented in section five. The proposed system is applied to a real construction project in section six, and the results of the structured interviews performed with the participants of the case project are presented in section seven. The limitations and advantages of the proposed system are revealed in sections eight and nine respectively. Finally, concluding remarks are made in section ten.

2. Literature review

Delay and withholding of payment is a serious problem in the construction industry globally [6,20]. Chan and Suen [21] identified payment issues as the number one source for the international project disputes in China. Arditi and Chotibhongs [22] performed a questionnaire survey to subcontractors, contractors, and employers in the United States, in which 89% of subcontractors claimed that the payments are delayed by more than 45 days after the completion of the work. Similarly, a report by the Australian Senate Economics References Committee reveals that contractors can employ tactics to subcontractors for accepting long payment claim periods, ranging anywhere between 30 and 90 days [23]. In-depth interviews with senior management personnel in a cross section of contractors, sub-contractors, suppliers, and consultants performed by the UK Office of Government Commerce indicated that payment delays up to 60 days for the payments from the employers were common [24]. Ramachandra and Rotimi [6] analyzed the payment losses and delays in New Zealand and made the conclusion that payment delays and losses are prevalent within the industry. Abdul-Rahman, Kho and Wang [20] focused on the underlying causes of late payment issues in Malaysia and revealed the cash flow problems due to employer's poor management as the most significant cause. Liu et al. [7] performed a comprehensive literature review on the dispute causes and determined payment delays as one of the top owner related causes.

Project Bank Accounts (PBA)s were recommended by the National Audit Office [25] to improve payment practices in UK, particularly to protect the payments to subcontracts. In September 2009, the Government Construction Board required Central Government Departments, their Agencies and Non-Departmental Public Bodies in the UK to use PBAs for at least 80% of the value of work sub-contracted, unless there are compelling reasons not to do so [9]. Despite the fact that PBAs offer major advantages for resolving some of the payment issues, there has been reluctance among employers and contractors for using PBAs, especially in the private sector [12,13]. Price [12] through a survey study, identified employer resistance to the initiative, burdens and costs of implementation, loss of cash flow benefits, staff training and company policy as the significant barriers and made the conclusion that the barriers are likely to prohibit the widespread implementation of PBAs in the near future. In a recent survey study, Griffiths, Lord, and Coggins [13] determined the main barriers for the use of PBAs as; resistance from main contractors, complex and confusing nature, and administrative demands. Macaulay and Summerell [10] mentioned the set-up and administration costs (in time and money) as the main disadvantage of PBAs.

Advances in blockchain technologies created smart contracts as a powerful alternative to develop a payment system for security of payment of construction contracts. The promise of smart contracts to protect main contractors, subcontractors and suppliers against the insolvency of the principal or late payments was initially mentioned by Cardeira [16]. Cardeira [16] also mentioned the need for stability and acceptance as the potential barriers for adoption smart contractors in the construction industry. Wang et al. [17] pointed out that with a smart contract, the funds or cryptocurrencies can be embedded into the contract against the insolvency of the late payments so as to eliminate the payment and cash-flow issues. Turk and Klinc [26] highlighted the potential of blockchain to make construction processes less centralized and pointed out the need for research in that direction. Mason [18] stated that the smart contracts could be extremely useful for collaborative construction and highlighted project bank accounts and project insurance as the potential areas, but also identified reliability and interoperability as temporary barriers for their adoption. Ahmadisheykhsarmast and Sonmez [19] mentioned the potential of smart contracts to reduce payment problems in the construction industry. Luo et al. [27] proposed a framework to demonstrate the potential of blockchain for automation of construction payments. An extensive review and analysis of the current state of blockchain in the construction sector are presented in Li et al. [28], in which three specific use cases are appraised for the construction industry, including PBAs as potential areas for blockchain. Hijazi et al. [29] discussed the advantages and limitations of integrating blockchain with BIM to enable a single source of truth. Li and Kassem [30] reported outcomes of semi-structured interviews to discover the potentials and challenges for blockchain and smart contracts in the construction industry in which enactment of new financial regulations is identified as a factor to overcome to implement smart contract cryptocurrency payments. Shojaei et al. [31] presented a smart contract implementation model for integrating BIM and blockchain. Hunhevicz and Hall [32] reviewed, analyzed, and categorized blockchain and smart contract use cases in construction literature and proposed a decision framework that can be implemented for designing construction blockchain and smart contract applications. Li et al. [33] proposed a blockchain and smart contract framework to increase traceability during operation of a built asset. Extensive reviews of the current status of construction blockchain research and use cases in the construction industry can be found in Li et al. [28] and Hunhevicz and Hall [32].

The potential of blockchain for the construction industry was mentioned in several research, however the use of blockchain in construction has been limited [26]. Wang et al. [17] provided an example smart contract in which if the temperature of the construction site is higher than a certain value, the owner pays an allowance to the contractor. Li et al. [34] developed a simulation which showed the functionality of automating the installation of an external thermal

insulation composite system in which the performance of the installation in the physical environment are processed through a code based on the agreements representing the contract, which accordingly authorizes or denies the payment. Elghaish et al. [35] presented a blockchain framework that for integrated project delivery, presented smart contract functions for financial transactions and examined the practicality of the built-up hyperledger network. Ahmadisheykhsarmast et al. [36] proposed a smart contract application for unguaranteed retention payment and documentation to present advantages and security issues of smart contracts for contract management.

Despite the fact that previous research has highlighted the importance of payment issues in the construction industry, very few proposed methods for eliminating or reducing the problem. The potential of smart contracts for construction industry was mentioned in several studies in recent years, however, very limited research has developed applications to explore their use in the construction industry. Hence, the main focus of this research is to narrow these gaps in the literature by designing and developing a smart contract system for security of payment of construction contracts.

3. Blockchain and smart contracts

Blockchain technology, pioneered by Bitcoin [37], provides a globally-consistent decentralized and distributed database that is formed of a growing list (chain) of records (blocks) which are secured using cryptographic proof instead of a central authority. Bitcoin presented an innovative decentralized electronic payment system that enabled any two parties to transact directly with each other without the need of a trusted third party. The cryptocurrency transactions are verified and added to the blockchain by a decentralized consensus process called mining. Bitcoin uses a mining procedure called Proof-of-Work (PoW). The idea of PoW was originally proposed by Dwork and Naor [38] to combat junk e-mail and controlling access to a shared resource. In PoW, the miners (network nodes that verify the transactions) compete to solve a moderately hard, but trivial-to-verify problem that was created by using a cryptographic hash function. The miner that wins the competition is rewarded with cryptocurrencies after 51% or more nodes verify the solution.

In recent years, Proof of Stake (PoS) based blockchain consensus algorithms were proposed as an efficient, fast, and inexpensive alternative. In PoS system, ownership of cryptocurrency of the blockchain is usually used as a proof of stake. A PoS blockchain agrees on new blocks through a process where each cryptocurrency holder can participate as a validator, according to the amount of the cryptocurrency it holds [39]. The validators that deposit their cryptocurrency for verification receive rewards such as earning interest.

Blockchain technology ensures that all the transactions cannot be changed or deleted and data is held on all of the nodes at the same time, hence the security of blockchain is ensured by distributed data which eliminates a single point of failure. In public blockchains, anyone can participate in decentralized consensus process and the blockchain is usually publicly broadcasted to ensure the transparency, accessibility, and stability of the transactions. In recent years private blockchain platforms have been proposed as an alternative to the public blockchain. In private blockchains, consensus process is achieved by a limited and permissioned participants and the blockchain is usually kept private. Hence, in general private blockchains are centralized and do not require decentralized mining consensus [40] and cryptocurrencies.

The introduction of smart contracts marked the start of blockchain 2.0 era in which execution of contract clauses can be performed on the blockchain with automated computerized protocols. The idea of embedding the contractual clauses into software as computerized transaction protocols was initially proposed by Szabo [41]. In addition to the capabilities of blockchain, smart contracts provide a secure and trustworthy platform for automated execution of contract clauses, without requiring a trusted intermediary such as lawyers or banks, hence resulting in savings in transaction fees, administrative costs and expediting the transaction process [42,43].

4. Research method

The research method for design and development of the proposed smart contract system consisted of seven stages. In the first stage, a general analysis is performed to define the requirements of the smart contract procedure for security of payment of construction contracts. The second stage included selection of the blockchain for the smart contract system for security of payment which was named SMTSEC. The third stage consisted of design of architecture of the SMTSEC, depicting the modules required and their relationships, and the choice of development platforms for the modules. In the fourth stage, the first module of SMTSEC which enables use of existing schedule and cost data is developed, and in the fifth stage the second module of SMTSEC, the smart contract-based decentralized application (DApp) is developed. The sixth stage consisted of testing of SMTSEC through a real construction project which is also used to illustrate the proposed procedure and to identify the advantages and limitations of the new smart contract system. Finally, in the seventh stage structured interviews were conducted to find out the reactions and opinions of the participants of the case project about SMTSEC. The research method of SMTSEC is summarized in Fig. 1. The details of the reseach methods including selection of the blockchain, design of architecture, development and coding, case project implementation, and structured interviews are provided in the following sections.

5. A smart contract system for security of payment of construction contracts

In this section the proposed smart contract procedure is described, followed by the design and development stages of the SMTSEC.

5.1. Smart contract procedure for security of payment of construction contracts

The proposed procedure is defined to enable security of payments for works under construction through smart contracts. In the procedure, the employer (EM) and the main contractor (MC) first agree on the terms of security of payment. The agreed terms are presented as a computerized protocol through a smart contract which is deployed on the blockchain as shown in Fig. 2. As in the majority of construction projects interim payments are made on monthly basis, the progress payments are assumed to be made monthly in SMTSEC. Once the project starts, the projected progress payment amount for the next progress payment period is determined using the planned cash flow of the project. The smart contract procedure enables blocking of the projected next month's payment at the beginning of the month to ensure the security of progress payments. Immediately after the employer's approval of the progress payment, smart contract automatically transfers the progress payment amount to the contractor's, subcontractors', and suppliers' wallets according to the agreed terms. The smart contract consecutively blocks the projected progress payment amount for the next period along with the transfer of the funds to secure the payments for the next payment period. The procedure is repeated for the next progress payment period until the project is completed.

Since the current commonly used decentralized smart contract platforms perform transactions with cryptocurrencies, the procedure requires use of a cryptocurrency in transactions. The smart contract conditions are unique for each project and should include, the fiat currency (Cur) in which the payments are to be made according to the contract, the smart contract cryptocurrency type (CryT) that the Cur will be converted, a contingency amount (Cong) which is the additional amount of CryT that will be blocked to manage possible CryT/Cur fluctuations and potential increases in the planned progress payments,

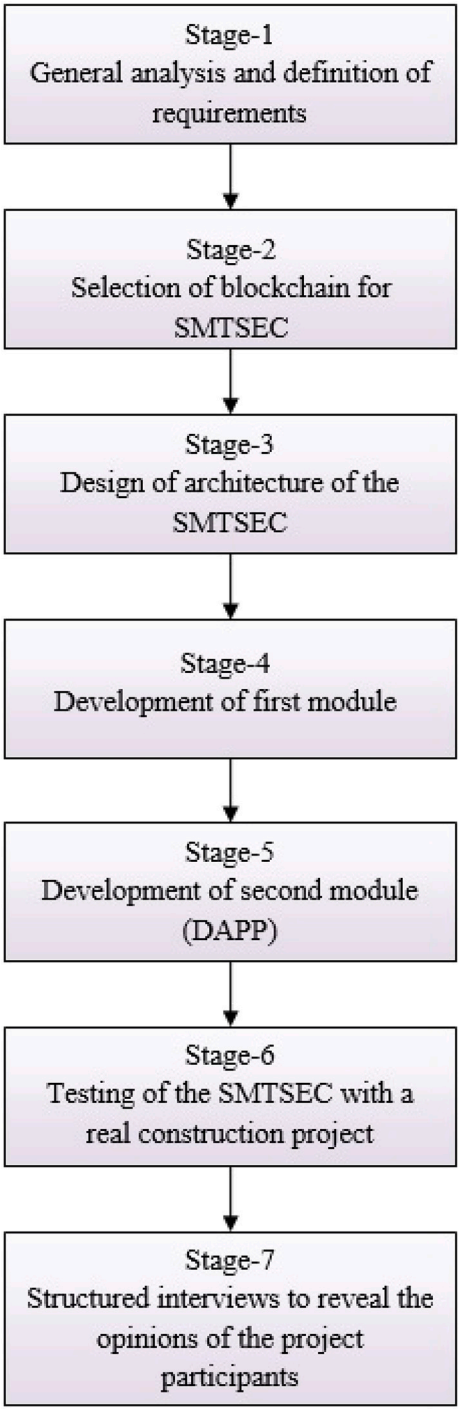


Fig. 1. Methodological steps of SMTSEC.

the period that the employer's funds will be blocked (PeriBloc) and cannot be used, the period that the employer's blocked funds can only be used for progress payments (PeriPay), and the percentage amount of payments that will be paid directly by the employer to the subcontractors and suppliers ($SubPer_i$). The employer's (WAdEM), contractor's (WAdMC), and subcontractors' and suppliers' ($WAdSC_i$) wallet addresses should also be included in the smart contract. The smart contract procedure enables releasing of the blocked amount if the contractor does not request a progress payment within PeriPay, or the employer does not approve the contractor's payment request within PeriPay to prevent the employer's funds to be blocked indefinitely in case of a dispute.

5.2. Selection of the blockchain for SMTSEC

Selection of an adequate blockchain is an important decision for the smart contract systems, which depends on numerous factors that are specific for the application. Li et al. [28] mentioned scalability, security, privacy, integration with hardware, and integration with software as the important factors for selection of the blockchain. Hunhevicz and Hall [32] identified throughput, data storage, interoperability, privacy, smart contracts, and cost structure as the most important technical considerations. In selection of the blockchain for the SMTSEC, blockchains supporting smart contracts are considered only, as the proposed system relies on smart contracts.

Ethereum, Hyperledger Fabric, Corda, Stellar, Rootstock, and EOS are identified as the major smart contract development platforms based on their popularity in developing community, and technical maturity [44]. The criteria transparency, maturity, popularity, security, size of market cap, and decentralization are considered for selection of the blockchain for SMTSEC among the major blockchains supporting smart contracts. Ethereum, Rootstock, and EOS are considered in the next stage, since they are public blockchains, and to achieve the objectives of transparency and decentralization for the proposed system public platforms are more suitable. The most commonly used blockchain among the blockchains supporting smart contracts is Ethereum [45], which also has the largest market cap among the public smart contract blockchains. The popularity of the Ethereum blockchain enables availability and diversity of the security enhancement alternatives, guides, resources, and tools for smart contract developers. The market cap of the blockchain is very important to provide the sufficient liquidity for progress payment transactions. Hence, Ethereum is selected as the blockchain for SMTSEC, however the proposed smart contract procedure can also be implemented on other smart contract platforms.

5.3. Design of SMTSEC

SMTSEC is designed to enable use of existing project schedule and cost data while performing the proposed smart contract procedure on the selected blockchain. Two modules are designed for SMTSEC. The first module consists of add-on software developed in Microsoft Project 2019. The objective of the first module is to transfer the required schedule and cost data to the smart contract through a project management software. Microsoft Project software is selected for the first module, as it is one of the most widely used software for project management [46], and it provides variety of tools and support for development of an add-on software. The second module consisted of a smart contract-based decentralized application which is designed to be deployed on the Ethereum blockchain. The DApp is composed of two parts a frontend, which enables the user interface, and a backend, which performs the smart contract procedure for security of payment of construction contracts. The frontend is also designed to interact with the smart contract deployed on the Ethereum blockchain. Solidity 0.4.0 language is selected as the smart contract language to develop the backend of the DApp. Solidity is chosen as the smart contract language, since it is the most well-known and mature smart contract-oriented language [47].

5.4. Microsoft Project add-on

The add-on module of SMTSEC is developed for Microsoft Project 2019 in C# with Visual Studio 2019. The first module named “MPP Parser” enables use of existing schedule and cost data approved by the employer to determine the projected and actual progress payment amounts and to facilitate the data exchange among the proposed system. The schedule and payment data including planned and actual completion dates, budgeted and actual payments, and the information of parties that will receive the direct payment from the employer according to the smart contract conditions should be included in a

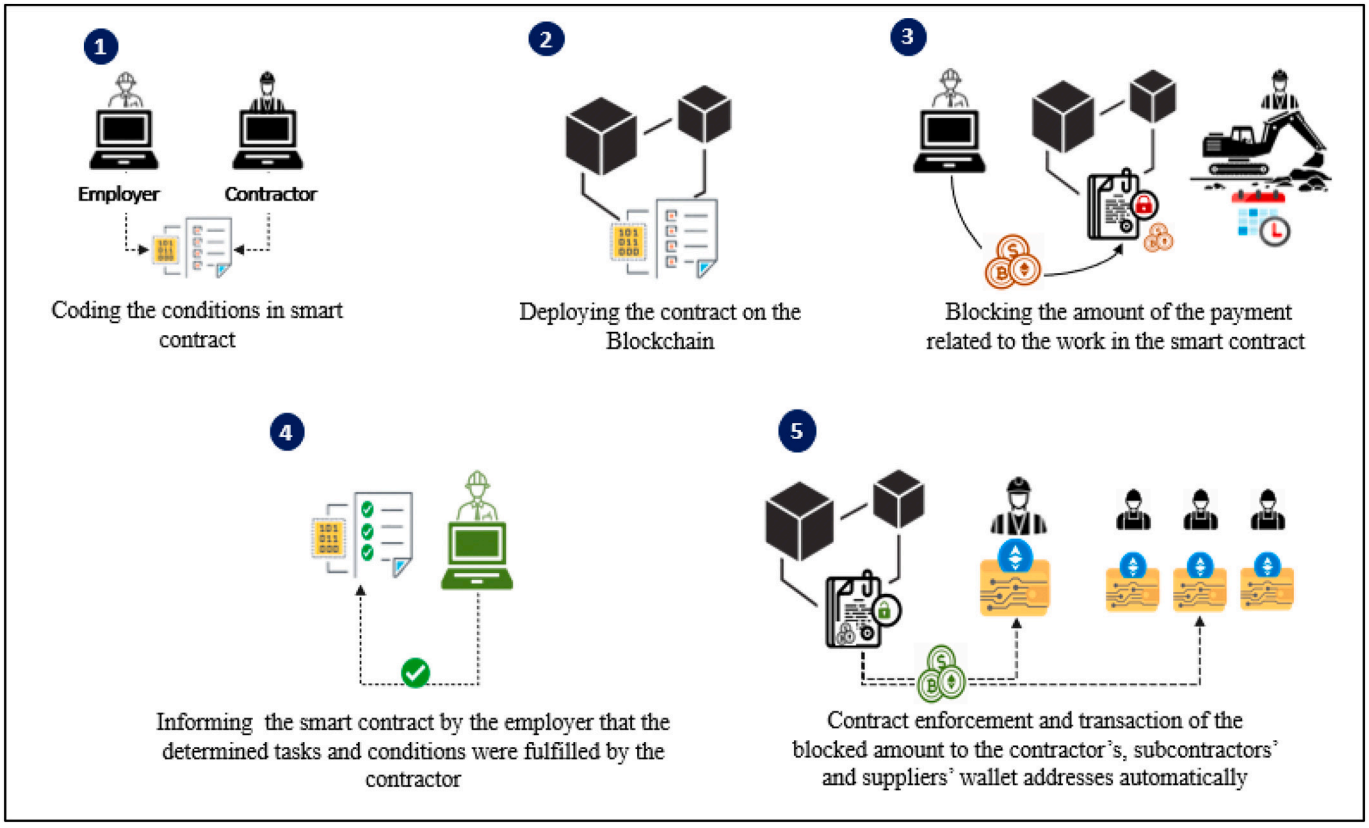


Fig. 2. Proposed smart contract procedure for security of payment of construction contracts.

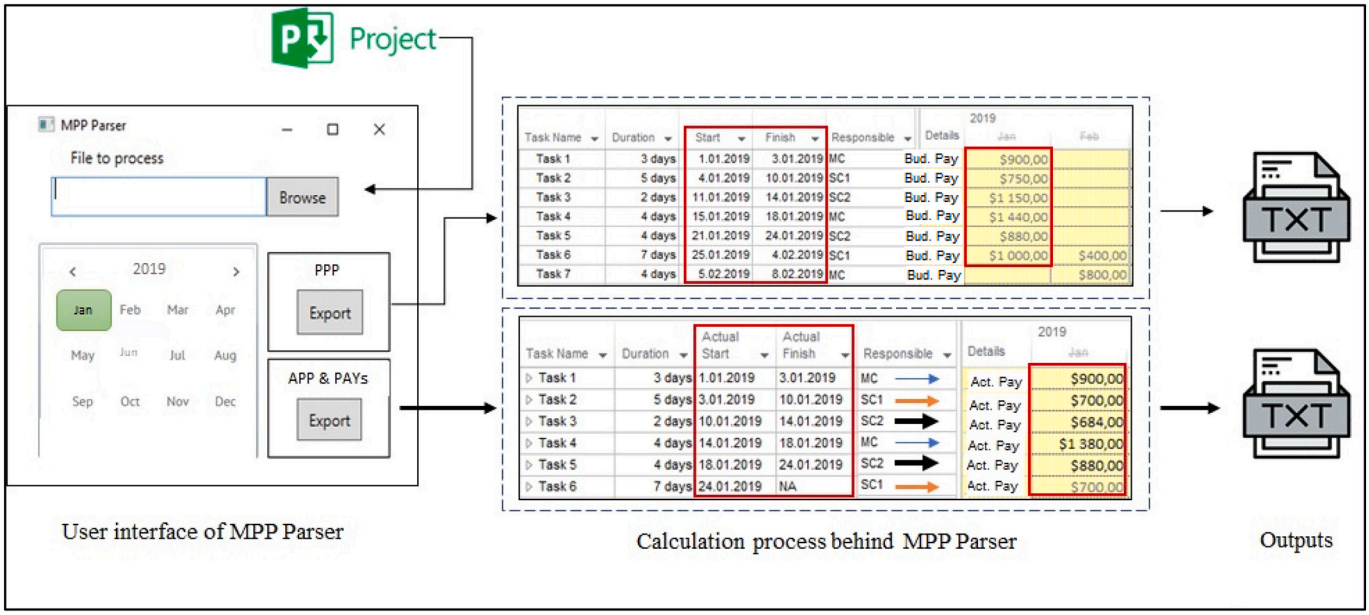


Fig. 3. MPP Parser module of SMTSEC.

Microsoft Project file to initiate the SMTSEC process. The month of the progress payment and the Microsoft Project file which includes the progress payment and schedule data should be selected by using the MPP Parser as shown in Fig. 3. At the beginning of the project, the contractor should select the first progress payment month and "PPP Export" button should be pressed to initiate blocking of the first month's projected progress payment amount (PPP). MPP Parser will create a ".TXT" file to transfer the projected progress payment amount to the second module of SMTSEC.

At the end of each month, the updated planned and actual schedule and payment data should be included in the Microsoft Project file to include the latest information in SMTSEC. MPP Parser will calculate the total actual progress payment amount (APP) and its breakdown for the selected month when "APP & PAYS Export" button is pressed. MPP Parser will create a ".TXT" file to transfer the actual progress payment amounts along with its breakdown to the second module of SMTSEC. MPP Parser provides a practical tool to deal with the dynamic nature of construction schedules.

5.5. Decentralized application

The second module of SMTSEC consisted of a DApp. The frontend of the DApp is developed by HTML5/CSS3/PHP/JavaScript. The smart contract part (backend) of the DApp is developed in Remix IDE with Solidity 0.4.0 language and Web3.js library is used for interacting HTTP web page with the blockchain node and the smart contract. The smart contract is deployed on Ethereum blockchain. The tests of SMTSEC were performed on the Ganache blockchain, which is a virtual Ethereum blockchain that does not require any deployment or transaction fees. CryT is taken as Ethereum (ETH) which is the cryptocurrency of the Ethereum blockchain, Cur is taken as the United States Dollar ($), and PeriBlock is taken as 30 days. However, other fiat currencies that can be converted in a cryptocurrency exchange can also be used in SMTSEC.

The blocking of the first month's PPP could be requested by the contractor right after the start of the project using the DApp. The DApp will get the PPP amount for the first month from the ".TXT" file, which was previously created by the contractor through the MPP Parser module. Once the DApp is executed, it will acquire the latest ETH/Cur exchange rate, and will then calculate the amount to be blocked (Bloc) by multiplying the PPP amount with the Cong and will display the Bloc for the first month in Cur and ETH on contractor's screen as shown in Fig. 4. The contractor should enter WAdMC's private key and press the "Blocking Request" button to complete the blocking request. The DApp will then show the request on the employer's screen under the "Amount to be Blocked" as shown in Fig. 5. The employer has to deposit the required funds in ETH to the WAdEM and then enter the private key of the WAdEM to approve the blocking of the funds. Once the funds are blocked, they cannot be used before PeriBloc is over. Besides, the DApp does not allow the contractor to claim the progress payment until PeriBloc is over. The pseudo code of blocking process is given in Fig. 6.

The progress payment of the first month could be requested by the contractor after PeriBloc is over. The DApp will display the APP as well as the progress payment amounts of the main contractor (PayMC), the subcontractors and suppliers (Pay_i) in Cur and ETH according to the updated data provided by the MPP Parser and the Microsoft Project file. The contractor should enter WAdMC's private key, and press the "Payment Request" button to send a request of payment to the employer as shown in Fig. 4, which will be displayed on the employer's screen under the "Actual Amount". The payment amounts will be automatically and simultaneously transferred to the main contractor's, subcontractors' and suppliers' specified wallets through the smart contract immediately after the employer's approval by entering the private key of the WAdEM. The excess amount of the blocked amount will be released and transferred to the WAdEM along with the payments. The DApp will alert the employer to deposit the deficit amount if the blocked amount is not sufficient to make the progress payment. The pseudo code of the payment process is provided in Fig. 7. Once the contractor, subcontractors, and suppliers receive the payments in ETH they can immediately convert it into any fiat currency in the local cryptocurrency exchanges. The blocking of the second month's payment can be requested by the contractor after the payments for the first month are made. The procedure is repeated every month until the project is completed. The flow chart of the blocking and payment processes is shown in Fig. 8.

SMTSEC enables the employer to release the blocked amount if the contractor does not request a progress payment within PeriPay, or the employer does not approve the contractor's payment request within PeriPay. Hence, the maximum period that the funds of the employer will be blocked is PeriBloc plus PeriPay. After the maximum blocking period, SMTSEC will enable releasing the blocked funds and will transfer them to WAdEM, if the employer presses "Withdraw Blocked Funds" button on the employer's screen. The release procedure prevents the employer's funds to be blocked by the smart contract indefinitely in case of a dispute. The pseudo code of the release process is given in Fig. 9.

6. Case project

SMTSEC is applied to a real construction project as a shadow system in parallel to the conventional payment system to illustrate the processes and benefits of the proposed system. The case project compromised civil works of a 3000 m^2 powerhouse building that is under construction in Turkey. The budgeted cost of civil works of the powerhouse building is $20 Million. There were two subcontractors; subcontractor-1 (SC1) is performing the reinforcement works, and subcontractor-2 (SC2) is carrying out the structural concrete works. The rest of the civil works were performed by the main contractor (MC). At the first stage, the information of Cong, PeriBloc, PeriPay, $SubPer_1$, $SubPer_2$, WAdEM, WAdMC, $WAdSC_1$, and $WAdSC_2$ were included in the smart contract, which is deployed on the Ethereum blockchain. In the

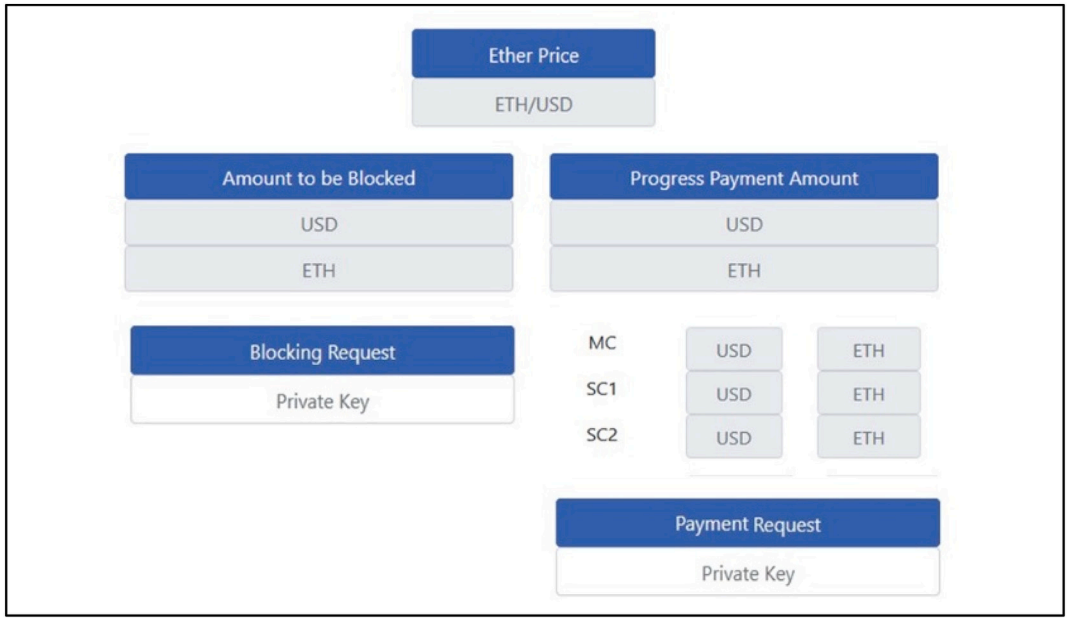


Fig. 4. Contractor's screen for the DApp module of SMTSEC.

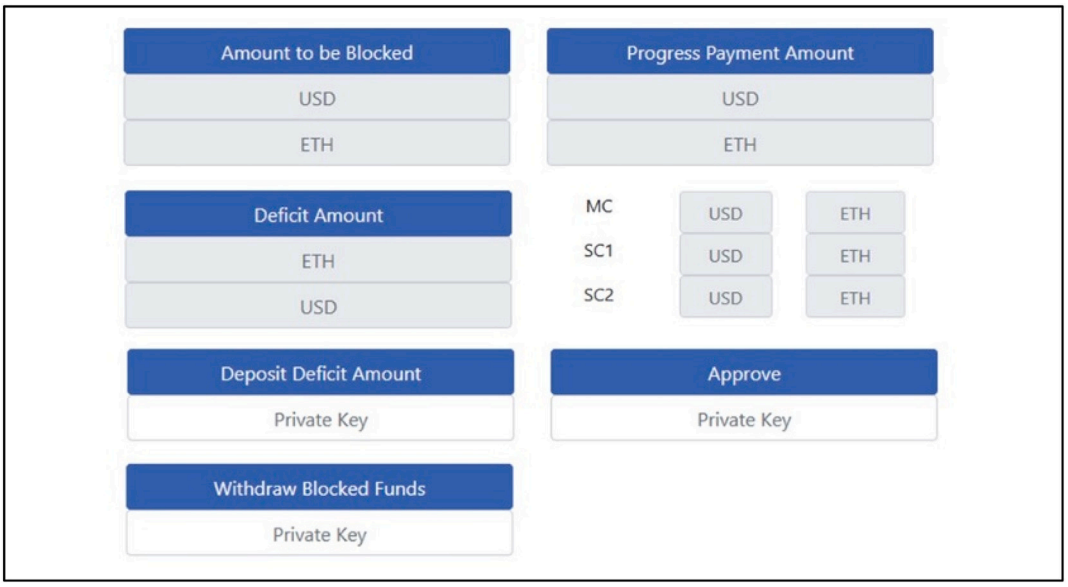


Fig. 5. Employer's screen for the DApp module of SMTSEC.

case study Cong was set as 20%, PeriBloc and PeriPay were taken as 30 and 60 days, respectively. $SubPer_1$ and $SubPer_2$ were both set as 50%, and the remaining payments to the subcontractors were paid by the main contractor after deduction of the overhead expenses, contractor's profit and subcontractor retentions.

The civil works of the case project started on March 1, 2019. MPP Parser module of SMTSEC was used to calculate the PPP amount for March 2019 and to create the ".TXT" file to transfer the first PPP amount to the DApp. The DApp was executed on March 1, 2019 to convert the first month's PPP of $272,417.74 to 1980.50 ETH at the exchange rate of 137.55 ETH/$. Hence, with a Cong of 20%, the Bloc was calculated as 2376.60 ETH by the DApp as shown in Table 1. The blocking request was submitted to the employer by entering the WAdMC's private key. Once the amount is approved by the employer, 2376.60 ETH was blocked by the smart contract and the balance of employer's wallet was reduced to 8664.34 ETH from 11,040.94 ETH as shown in Fig. 10.

The request for the first progress payment was made on April 1, 2019. The actual progress data of the contractor at the end of March 31, 2019 was used to calculate the progress payment amounts for the MC, SC1, and SC2. The progress payment amounts of MC, SC1, and SC2 were calculated as $144,897.92, $52,334.95 and $47,277.50 respectively by MPP Parser based on the progress values given in Fig. 11. The payments amounts for the MC, SC1, and SC2 were calculated as $194,704.15, $26,167.48 and $23,638.75 respectively, according to the $SubPer_1$, $SubPer_2$ amounts specified in the smart contract. The fiat currency payment amounts were converted as 1369.23 ETH, 184.02 ETH, and 166.24 ETH respectively by the DApp module of SMTSEC at exchange rate of 142.20 ETH/$ for April 1, 2019, and the progress payment request was made. The amounts were transferred to the WAdMC, $WAdSC_1$, and $WAdSC_2$ by the smart contract immediately after employer's approval, since the blocked amount of 2376.60 ETH was larger than the actual total progress payment amount of 1719.48 ETH. The excess amount of 657.12 ETH was released and was transferred to the WAdEM by the smart contract, increasing the WAdEM's balance to 9321.46 ETH as shown in Fig. 12. The transactions on the Ganache blockchain were performed within seconds. On the Ethereum blockchain standard transactions take between 15 s and 5 min [48] with the current PoW protocol. Once the main contractor and subcontractors receive the payment amounts in ETH they can convert it to any fiat currency in the local cryptocurrency exchanges.

The deployment cost of the smart contract for the case project was 0.0400752 ETH or $5.51 at an exchange rate of 137.55 ETH/$. There will be also be a development cost if the employers want to develop their own DApp. However, the development costs of the proposed DApp is a minor cost for most of the employers, since only a few basic clauses are included in the proposed smart contract system.

```
function PPP Blocking {
    require MC's claim through SMTSEC ;
    Retrieve the PPP amount from MS Project cashflow;
    Convert the fiat currency (Cur) of the PPP amount into the CryT;
    Bloc = (Cong) x (PPP)
    require EM's approvement  through SMTSEC;
    if WAdEM's balance ≥  Bloc Then
        Block  Bloc for the PeriBloc = true;
    else
        Alert the EM through SMTSEC to deposit the deficit (Bloc - WAdEM's balance) ;
    end;
}
```

Fig. 6. Pseudo code of blocking process.

```
function Payment {
    require MC's claim through SMTSEC;
    Retrieve the APP amount from MS Project;
    Convert the fiat currency (Cur) of the APP amount into the CryT;
        if PeriBloc period passed;
        require MC's claim through SMTSEC;
        require EM's approvement through SMTSEC;
                        if APP = Bloc Then
                            Unblock Bloc = true;
                            Transfer the PayMC of the MC to its WAdMC;
                            Transfer the Pay_i of the SC_i to their WAdSC_i according to the SubPer_i;
                        elseif APP < Bloc Then
                            Unblock Bloc = true;
                            Transfer the PayMC of the MC to its WAdMC;
                            Transfer the Pay_i of the SC_i to their WAdSC_i according to the SubPer_i;
                            Transfer the excess amount (Bloc – APP) to WAdEM;
                        else
                             Do not unblock Bloc = true;
                            Alert the EM to deposit the deficit amount (APP-  Bloc) through SMTSEC;
                         end;
    else;
           Alert the MC through SMTSEC that the APP could not be claimed before PeriBloc is over;
    end;
    }
```

Fig. 7. Pseudo code of the payment process.

7. Interviews with the projects participants

Structured interviews were conducted to reveal reactions and opinions of the parties involved in the case project about the proposed smart contract system. Structured interviews sometimes also referred to as standardized interviews, are universally used in social sciences in which the questions, in many cases the answer categories, have been fully developed, and the questions are put in an interview schedule before the interview begins [49]. The following structured interview procedure described in Fowler [49] is implemented in the interviews:

1) The questions were asked exactly as they are worded and asked in the pre-defined order according to the interview schedule.
2) If an interviewee failed to answer a question completely, follow-up questions are asked.
3) The answers were recorded without discretion by the interviewer.
4) The interviewer minimized personal judgement and feedback to achieve accurate answers.

The interviewees consisted of all of the participants of the case project that had a position related to the progress payments and that wanted to participate in the interview. Hence, three project participants from the employer organization, three from the contractor, and two participants from the two subcontractor organizations were interviewed. Four interviewees were familiar with cryptocurrencies, however only one interviewee had previous knowledge of smart contracts before SMTSEC as shown in Table 2. A brief 10 min introduction to smart contracts was provided to the seven interviewees that are not familiar with the subject before the interview so that they can answer the questions comfortably.

The interviewees were asked four pre-worded questions in the same order. The questions included the advantages and disadvantages of the SMTSEC, their opinion on using SMTSEC for security of payment of construction contracts, and their suggestions about the SMTSEC. The comments of the interviewees included specific comments about the SMTSEC along with the general comments about the smart contracts, or the features of project bank accounts, such as employer's direct payments to the subcontractors, or the security of payment of construction contracts, which are not familiar to majority of the Turkish employer, contractor or subcontractor organizations. The questions asked in the interview and the answers of the interviewees are given in Table 3 to Table 6.

Seven categories are used to classify the responses to the first question on the advantages of SMTSEC, which are: 1) Improved financial management, 2) Elimination of payment issues, 3) Reduction of bid amounts, 4) Improved project performance, 5) Reduction of advance payment, 6) Improved trust, and 7) Increased motivation. The advantages of the SMTSEC in terms of perspectives of parties involved in the case project are provided in Table 3. All of the eight interviewees mentioned the potential of SMTSEC to eliminate or reduce the payment issues. Three interviewees (one from employer, one from the contractor, and one from the subcontractor) also pointed out that elimination of the payment issues would improve the project performance. Two participants from the employer identified improved financial management as one of the advantages of the SMTSEC, as it enforces financial management ahead of time. One interviewee (employer) commented that SMTSEC might decrease the bid amounts of the contractors as it secures their payments. Similarly, another interviewee (employer) mentioned that SMTSEC might lead to reduction of the advance payment. Increased motivation was mentioned as a potential by one participant (contractor).

The responses to the second question, which included disadvantages of SMTSEC, are classified in eight categories as: 8) Disputes, 9) Lack of legal infrastructure, 10) Loss of interest payments, 11) Blockchain risks, 12) Lack of tax procedures, 13) Privacy issues, 14) Loss of authority, 15) Payment amounts to the subcontractors. The perceived disadvantages of the SMTSEC along with their classifications are given in Table 4. One interviewee from the employer organization and one interviewee of the

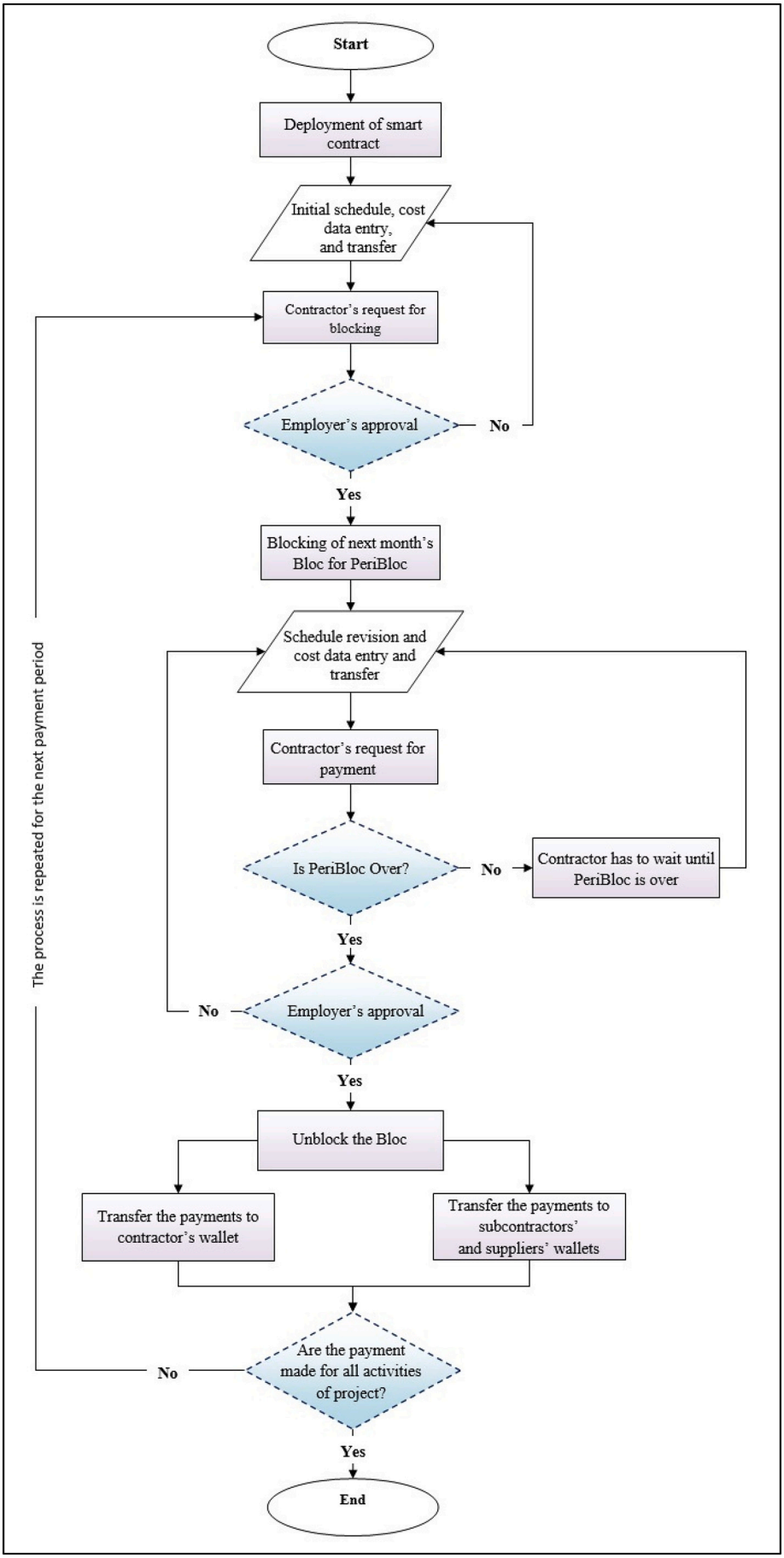


Fig. 8. Flow chart of blocking and payment processes.

```
function Release of Bloc {
    require EM's request through SMTSEC
        if PeriBloc + PeriPay period passed Then
            require EM's Private key;
             Transfer the Bloc to WAdEM;
        else
              Alert the EM that Bloc could not be withdrawn ;
         end;
}
```

Fig. 9. Pseudo code of the release process.

Table 1
Calculation of bloc.

First month's PPP ($)	272,417.74
ETH/$	137.55
First month's PPP (ETH)	1980.50
First month's PPP with %20 Cong (ETH)	2376.60
WAdEM's balance before Blocking	11,040.94
WAdEM's balance after Blocking	8664.34

contractor organization pointed out that direct payments made to the subcontractors by the employer might open the door for the employer to be a part of a potential payment dispute between the contractor and a subcontractor, which may complicate the resolution of the dispute. Lack of legal infrastructure for the smart contracts was mentioned by the two participants (employer). Two interviewees (contractor) emphasized privacy issues and stated that the information of payments made to the subcontractors should be kept private and should not be made available to the employer. The possibility of loss of authority and loss of control of the subcontractors was highlighted as a disadvantage by the two participants of the contractor organization. A potential blockchain outage was identified as a risk by one participant (employer). One interviewee mentioned loss of interest, and another (employer) lack of taxing and accounting procedures for cryptocurrency payments (employer) as the disadvantages. A respondent (subcontractor) stated that SMTSEC enables employer only to pay a portion of their payment, and the remaining part may still be subject to payment issues.

Three interviewees (two subcontractor, one contractor) said that they would use the SMTSEC without a condition as shown in Table 5. One respondent (employer) opposed using SMTSEC as it blocks the employer's funds. Four interviewees (two employer, two contractor) had reservations in using the SMTSEC. Two respondents from the contractor organization stated that they would use SMTSEC only after it is modified to require the contractor's approval for the subcontractor payments made by the employer. One interviewee (employer) mentioned that he would use SMTSEC after it is fully tested, another (employer) said he might use after the required legal infrastructure is ready. Two interviewees from the employer organization suggested that the provisions for damages (such as delay penalty) should also be included in the system as shown in Table 6. One respondent (employer) commented that the system might share information with the relevant third parties.

8. Limitations of SMTSEC

SMTSEC provides a decentralized, secure, efficient, transparent and trustworthy alternative for security of payment of construction contracts. Despite its advantages, SMTSEC has few limitations. CryT/Cur fluctuations is one of the main limitations of the proposed system. Since the Bloc in CryT will be blocked for PeriBloc, the employer may have losses if CryT depreciates against Cur during PeriBloc. Currently, there

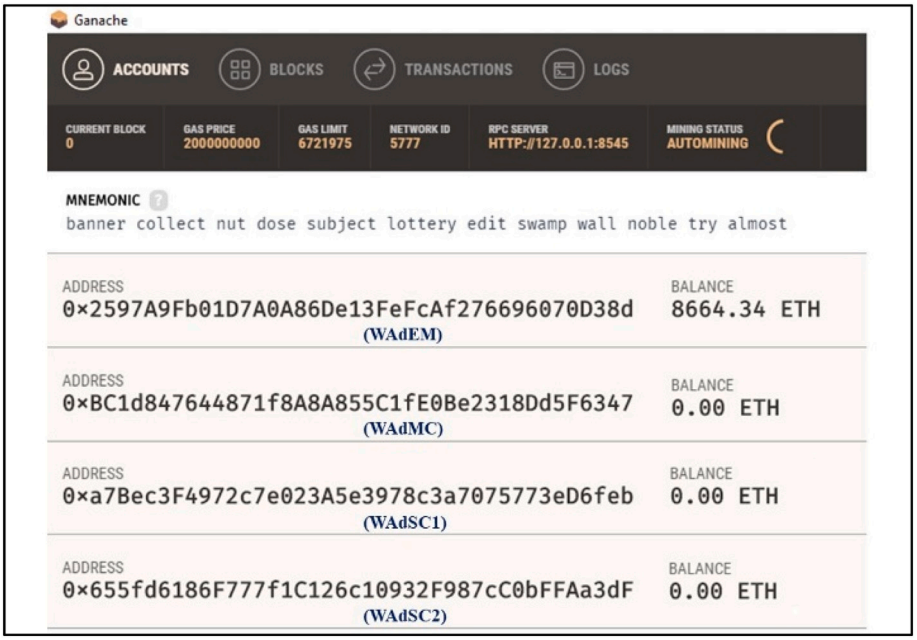
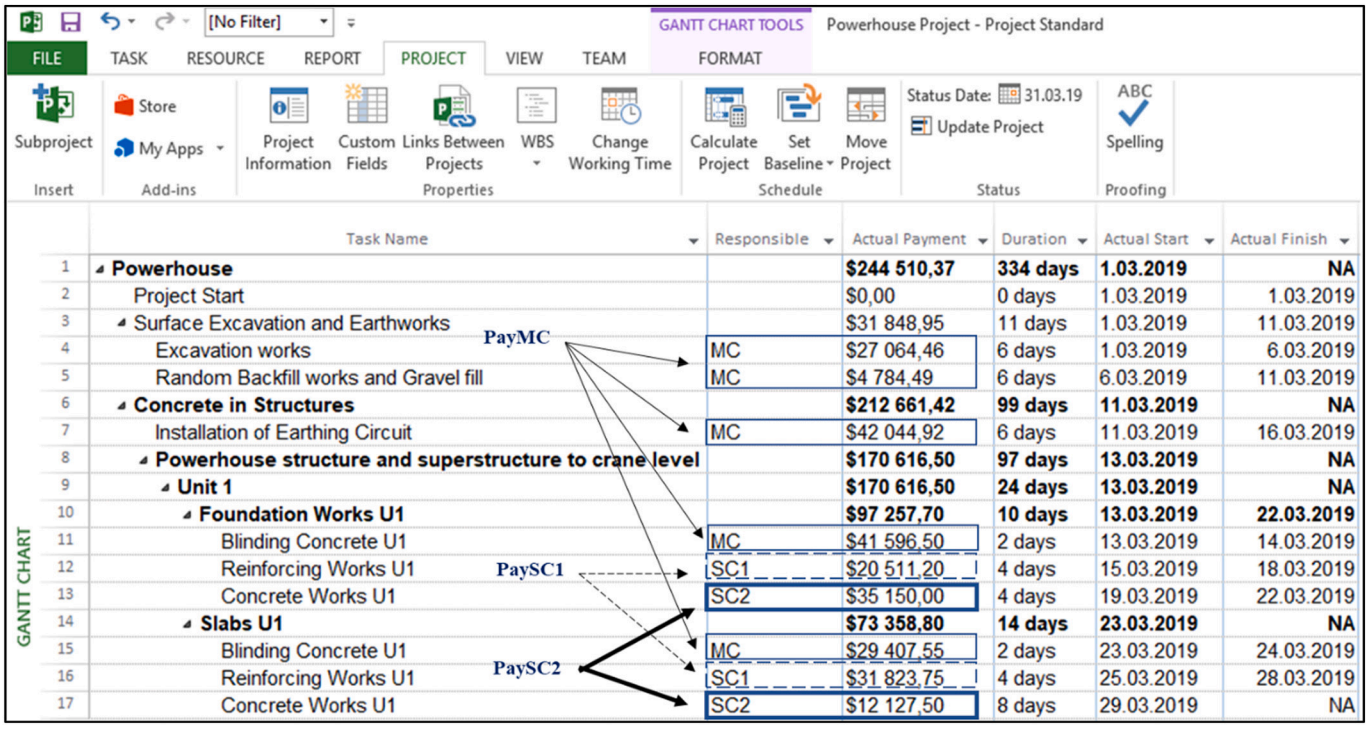


Fig. 10. Balances of wallets for the case project after blocking process.



	Task Name	Responsible	Actual Payment	Duration	Actual Start	Actual Finish
1	**Powerhouse**		**$244 510,37**	**334 days**	**1.03.2019**	**NA**
2	Project Start		$0,00	0 days	1.03.2019	1.03.2019
3	Surface Excavation and Earthworks		$31 848,95	11 days	1.03.2019	11.03.2019
4	Excavation works	MC	$27 064,46	6 days	1.03.2019	6.03.2019
5	Random Backfill works and Gravel fill	MC	$4 784,49	6 days	6.03.2019	11.03.2019
6	**Concrete in Structures**		**$212 661,42**	**99 days**	**11.03.2019**	**NA**
7	Installation of Earthing Circuit	MC	$42 044,92	6 days	11.03.2019	16.03.2019
8	**Powerhouse structure and superstructure to crane level**		**$170 616,50**	**97 days**	**13.03.2019**	**NA**
9	**Unit 1**		**$170 616,50**	**24 days**	**13.03.2019**	**NA**
10	**Foundation Works U1**		**$97 257,70**	**10 days**	**13.03.2019**	**22.03.2019**
11	Blinding Concrete U1	MC	$41 596,50	2 days	13.03.2019	14.03.2019
12	Reinforcing Works U1	SC1	$20 511,20	4 days	15.03.2019	18.03.2019
13	Concrete Works U1	SC2	$35 150,00	4 days	19.03.2019	22.03.2019
14	**Slabs U1**		**$73 358,80**	**14 days**	**23.03.2019**	**NA**
15	Blinding Concrete U1	MC	$29 407,55	2 days	23.03.2019	24.03.2019
16	Reinforcing Works U1	SC1	$31 823,75	4 days	25.03.2019	28.03.2019
17	Concrete Works U1	SC2	$12 127,50	8 days	29.03.2019	NA

Fig. 11. Actual progress of the case project at the end of March 312,019.

are very limited options to hedge the CryT/Cur fluctuations. Bitcoin futures which will be provided by the Chicago Mercantile Exchange (CME) enables limited hedge of the risks, as there may not always be a positive correlation with the Bitcoin and CryT prices. Ethereum futures which might be also a possibility in the near future would provide a better hedge; however, both of these alternatives come at a cost. A future widely accepted smart contract stable cryptocurrency, which has always a fixed value against a fiat currency would minimize the CryT/Cur fluctuations risks.

The current market cap of smart contract cryptocurrencies also presents a limitation on the use of smart contract-based security of payment systems for large size construction projects. With the current supply of 108,197,362 [50] and at a current exchange rate of 172.53 ETH/$, the current market cap of ETH is at $18.667 Billion. A large amount of buying demand from an employer for a mega project may lead to rapid appreciation of ETH against Cur, similarly large amount of selling orders from the contractors, subcontractors, and suppliers may result in a rapid depreciation of ETH against Cur. Hence, the proposed SMTSEC is more suitable for small or medium size construction projects.

Loss of interest payments is another limitation of the SMTSEC, as the blocked funds will not earn an interest, which is possible for PBAs.

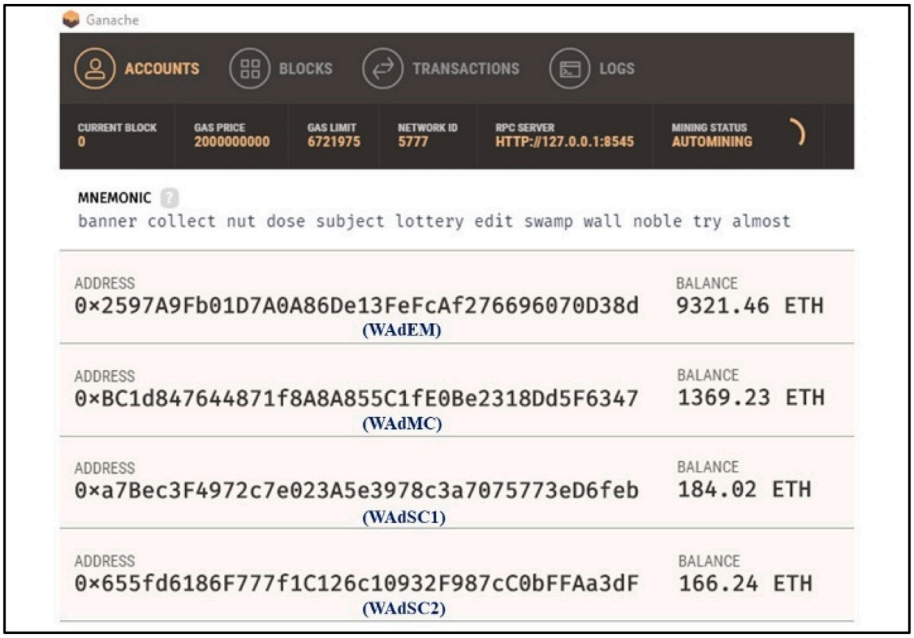


Fig. 12. Balances of wallets for the case project after payment process.

Table 2
Information of interviewees.

ID	Organization	Position	Level of knowledge	
			Cryptocurrencies	Smart Contracts
EM1	Employer	Project Manager	Low	None
EM2	Employer	Planning Engineer	None	None
EM3	Employer	Project Controls Engineer	Low	None
MC1	Main Contractor	Construction Manager	Low	None
MC2	Main Contractor	Project Controls Manager	None	None
MC3	Main Contractor	Contract Manager	Medium	Medium
SC1	Subcontractor	Company Owner	None	None
SC2	Subcontractor	Company Owner	None	None

However, when Ethereum blockchain moves to the PoS system it will enable the validators to earn interest [51], hence earning interest might be a possibility with the SMTSEC in the future. Despite the fact that blockchain and smart contracts provide a very secure platform for the development of decentralized applications such as SMTSEC, they are not risk free. Li et al. [45] provide a systematic review of security threats of the blockchain and smart contract systems and present enhancements to minimize the security risks.

The interviewees that evaluated the proposed smart contract system were limited to the participants of the case project. None of the participants that were included in the evaluation of SMTSEC reported any usability issues, however lack of usability testing is another limitation of the proposed system. A more comprehensive evaluation including a large sample of participants and additional real-life case projects would provide an in-depth assessment of SMTSEC.

9. Discussion

SMTSEC revealed the potential of smart contracts to protect main contractors, subcontractors and suppliers against the insolvency of the principal or late payments which was mentioned in Cardeira [16], Wang et al. [17], and Ahmadisheykhsarmast and Sonmez [19]. SMTSEC guarantees availability of the funds for a progress payment period by blocking the projected progress payment amount at the beginning of the progress payment period. SMTSEC allows direct payment from the employer's wallet to the subcontractors' and suppliers' wallets for improving cash flows and reducing payment issues. SMTSEC also provides transparency as the contractors, subcontractors, and suppliers can track all the transactions made from the employer's wallet. SMTSEC allows flexibility for security of payment conditions by presenting the parameters; Cong, PeriBloc, PeriPay, and $SubPer_i$, which could be easily included in the smart contract according to the agreements made between the employer and the main contractor. Hence, SMTSEC provides an alternative for eliminating or reducing the main disadvantages of PBAs including; complex and confusing nature, administrative demands [13] and set-up and administration costs [10]. The proposed smart contract system contributes to the literature and practice not only by presenting an alternative for the PBAs to resolve the payment issues of construction projects, but also by revealing the potential of smart contracts for the construction sector.

10. Conclusions

This paper presents a new smart contract payment security system for eliminating or reducing payment issues in the construction sector. The smart contract-based system is designed and developed to guarantee security of payments of works under construction and to present a platform for secure, efficient, timely and transparent payment of construction projects. The potential advantages and limitations of the proposed system were revealed through a real construction project.

The main contribution of the proposed smart contract payment security system is that it provides a secure, efficient, and trustworthy platform for security of payments of construction contracts, without requiring a trusted intermediary such as lawyers or banks. Hence, it has a potential to enable savings in the fees, reduce administrative costs and burdens, and expedite the payment process. The presented process of blocking of the payments of works that will be constructed for the next progress payment period enforces the employer to plan and arrange the payments ahead of time, and guarantees timely payment of progress payments to the main contractor, subcontractors, and suppliers. The proposed system promotes transparency for building trust among the project participants, by enabling the main contractor, subcontractor, and suppliers to view the dates and amounts of the all payments made by the employer,

The proposed smart contract payment security system receives the schedule and cost data through a project management software, hence it does not present an automated progress payment system. Future research focusing on integration of building information modeling to the smart contracts could enable automated smart contract progress payment systems. Building information modeling presents a major potential for representing the construction works as digital objects in the smart contracts. Smart contract systems for security of payment of procurement and guarantee of performance could be other promising areas for future research.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to

Table 3
Question 1: What are the advantages of SMTSEC?

Interviewee	Response	Category
EM1	Enables a system for realistic and organized cash flow management for the employer, hence has a potential to contribute to the employer's financial management.	1
EM2	May minimize the payment issues, which could decrease the bid amounts of contractors.	2,3
	May prevent halting the activities performed by the subcontractors due to payment problems.	4
	Improves financial planning and management.	1
EM3	Has a potential to eliminate majority of the current payment issues of the construction industry.	2
	Could improve the trust between the employer and the contractor.	5
	May decrease the amount of advance payment and advance payment bond expenses.	6
MC1	The contractor could concentrate to work better since he will not worry about the payments.	2,4
	Getting the payments on time will increase the motivation of the contractor.	7
MC2	Would improve the cash flow of the contractor.	2
MC3	As the contractor receives the payments of time, he could pay all his bills on time.	2
SC1	If the subcontractors receive their payments on time, they could also make their payments on time.	2
SC2	When all the payments are made on time the project performance could improve substantially.	2, 4

Table 4
Question 2: What are the disadvantages of SMTSEC?

Interviewee	Response	Category
EM1	The employer in general does not pay the subcontractors, if there is a dispute between the contractor and a subcontractor this system may also require the employer to be a part of the dispute.	8
EM2	I don't think an employer will have the progress payment amounts blocked without having a legal document.	9
	The employer will not be able to use the blocked amount, will lose the interest.	10
EM3	If a blockchain outage occurs not only the employer's blocked amount could be lost but also all the records of the payments may not be recovered. (I know this is a very low probability event but the impact is very high).	11
	The legal infrastructure for the system is not ready yet. If there is a dispute about the smart contract in my opinion currently there is not a court or an arbitrator that can resolve the conflict.	9
	There are also no tax procedures on payments made by cryptocurrencies.	12
MC1	The payments made by the contractor to the subcontractors is usually private information, the contractors do not want the employer to know this information.	13
MC2	I do not like the idea of employer making payments to the subcontractors instead of the contractor. The contractors want to manage the payments to the subs and they do no not want the employer to have any information about the subcontractor payments.	13
	In this system the employer may involve with the payment disputes between the subcontractors and the contractor, which may complicate the resolution of the dispute.	8
	If the subcontractors think that their payments are guaranteed, the contractor may lose the control and the subcontractors may relax and may underperform.	14
MC3	The employer manages part of the subcontractor payments hence the contractor may lose control of the subcontractors.	14
SC1	In my opinion there are no disadvantages.	–
SC2	The employer only pays part of the subcontractor payments, there still could be problem on the remaining part.	15

Table 5
Question 3: Would you use SMTSEC? Why/Why not?

Interviewee	Response	Category
EM1	Yes, as it could improve the chance of project success. However, I would not be the first one that uses the SMTSEC. I would wait until I am sure that the system is secure and does not present any problems that could not be handled.	Conditionally
EM2	No, the employer's progress payment amounts should not be blocked.	No
EM3	I am not sure, may be after the legal infrastructure for the system is ready.	Conditionally
MC1	If the contractor approves the payments of the subcontractors before they are paid, I would use the system.	Conditionally
MC2	Yes I would, but the subcontractor payments should be approved by the contractor.	Conditionally
MC3	Yes.	Yes
SC1	Definitely, Yes.	Yes
SC2	Yes, we have payment problems all the time, the system would guarantee timely payment of at least a portion of the payment which is very important for the subcontractors.	Yes

Table 6
Question 4: Do you have other suggestions to improve the SMTSEC?

Interviewee	Response	Category
EM1	The system may share information with the third parties (such as the engineer or the lawyers).	16
EM2	The penalties that are included in the contract should also be included in the system.	17
EM3	If penalties are also included in the system (such as delay penalty) the system could be used more effectively.	17
MC1	The system should enable the contractor to approve the progress payments of the subcontractor before they are paid by the employer.	14
MC2	The contractor should approve all of the subcontractor payments.	14
MC3	I do not have any suggestions.	–
SC1	This is a very useful system for us to receive the payments on time.	2
SC2	The system should make full payment to the subcontractors instead of partial payment.	15

influence the work reported in this paper.

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